# Utilizing Explainable AI to Enhance Real-time Student Performance Prediction in Educational Serious Games

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## ABSTRACT

Serious Games (SGs) enable the collection of valuable student interaction data, allowing for the analysis of student performance through Artificial Intelligence (AI). However, the lack of explainability in AI models represents a significant challenge. This research aims to create an interpretable model capable of predicting students' real-time performance while playing a SG. We developed a model for task completion prediction, resulting in 77.21% balanced accuracy. Additionally, the incorporation of eXplainable AI (XAI) ensures interpretability, supporting personalized learning experiences and unlocking AI benefits for non-technical users.

## CCS CONCEPTS

• **Computing methodologies → Machine learning approaches**;
• **Applied computing → Computer-assisted instruction**;

## KEYWORDS

Learning analytics, artificial intelligence, machine learning, xAI

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## 1 INTRODUCTION

In recent years, the use of games in educational contexts has experienced a remarkable increase. Specifically, Serious Games (SGs) – games that do not have entertainment, enjoyment, or fun as their primary purpose [3] – can provide an excellent context that facilitates the acquisition and assessment of knowledge and skills [1]. However, real-time monitoring of students creates challenges for educators, addressed through Artificial Intelligence (AI) and Machine Learning (ML) models, capable of handling large amounts of information and recognizing behavioral patterns. Moreover, to address the lack of explainability in AI models, eXplainable AI (XAI) proposes a suite of techniques that aim to produce more explainable models without compromising performance.

This research aims to create an interpretable model for real-time prediction of students' outcomes in a SG. We designed a set of features derived from *Shadowspect*, a game designed as a formative assessment tool to assess mathematical content standards. Additionally, we developed a set of ML models to predict performance in real time while improving explainability. We found previous studies [2, 4] that used ML models to analyze students' interaction and predict their performance using in-game features. Specifically, Jang et al. [2] used several ML models to predict performance and verify whether at-risk students could be distinguished, showing helping information to each student using XAI techniques. Our research stands out from existing literature on predicting student performance in SGs by addressing two crucial aspects: real-time prediction and model explainability.

## 2 CONTEXT AND FEATURE ENGINEERING

In our research, we have used *Shadowspect*, a 3D geometry game specifically designed to evaluate math core standards. For the data collection process, the team recruited seven teachers who used the game with students from seventh to tenth grade. The final dataset used for this research consists of approximately 428,000 events performed by a total of 322 students (with an average of 1,320 events per student). We employed a multi-prediction approach based on distinct time intervals derived from the average time required to complete each in-game puzzle. These intervals are set at 25%, 50%, and 75% of the average completion time for each puzzle.

We designed a set of features crucial for predicting user success, categorized into three groups. **User features** provide insights into the user's overall performance and interaction patterns. These features include: `percentage_tutorial` (percentage of completed puzzles in the tutorial category), `percentage_intermediate` (percentage of completed puzzles in the intermediate category), `percentage_advanced` (percentage of completed puzzles in the advanced category), `attempts_per_puzzle` (average number of solution attempts needed by the user to solve a puzzle), and `user_elo`

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(Elo ranking [5] obtained by facing each user against different puzzles.)

**Puzzle features** cover a set of data from the different levels within the game, allowing us to capture the unique characteristics and difficulty of each level. These features include: `puzzle_difficulty` (aggregated measure calculated by using the average time it takes to complete the puzzle, the average number of actions to solve it, the percentage of abandonments, and the percentage of incorrect checks when attempting to solve it), `puzzle_elo` (Elo ranking [5] obtained by facing each puzzle against different users), and `puzzle` (the puzzle identifier).

Finally, **attempt features** provide specific information about what the user is doing in that specific attempt, giving us a detailed view of the actions and decisions during gameplay. These features include: `n_events` (total number of events generated by the user), `n_breaks` (number of user's idle moments), `n_snapshot` (number of screenshots taken by the user), `n_rotate_view` (number of camera rotations), `n_manipulation_events`, `n_check_solution` (number of attempts to check the solution), and `best_submit` (determined by dividing the number of matched silhouettes by the total number of correct silhouettes).

It is important to note that **puzzle features** and **user features** do not vary with different time intervals since they aim to provide an overview of the user and the puzzle. Therefore, features varying with time intervals will be **attempt features**, which provide information about the current game session.

## 3 EXPLAINABLE MODEL AND CONCLUSIONS

Regarding the data preprocessing, we randomly selected 70% of users as training users and the remaining 30% as testing users. In addition, we applied a one-hot encoding of the `puzzle` feature because some of the ML models used are unable to handle categorical data. For model training and configuration we considered a set of seven algorithms, we employed ten-fold cross-validation, and we used balanced accuracy as the performance metric, which is highly useful in scenarios for imbalanced data. The best results obtained by each algorithm and time interval in the cross-validations are summarized in Table 1. The table shows that RF achieved the best performance in every time interval, achieving a balanced accuracy of 0.761 in the first time interval (25%), 0.778 in the second time interval (50%), and 0.792 in the third time interval (75%).

| Model | SGD | KNN | DT | RF | AdaBoost | MLP | SVC |
|---|---|---|---|---|---|---|---|
| **Prediction 1 (25th)** | 0.7229 | 0.6744 | 0.7448 | **0.7614** | 0.7253 | 0.7407 | 0.6964 |
| **Prediction 2 (50th)** | 0.7096 | 0.7000 | 0.7433 | **0.7783** | 0.7620 | 0.7652 | 0.7193 |
| **Prediction 3 (75th)** | 0.7035 | 0.7217 | 0.7564 | **0.7922** | 0.7789 | 0.7884 | 0.7353 |

**Table 1: Ten-fold cross-validation training results.**

After selecting RF as the best algorithm, we evaluated the generalization power of the three RF models (one for each time interval) on the test set, which contained unseen data. The models achieved a balanced accuracy of **0.772** in the first time interval, **0.792** in the second, and **0.793** in the third. Due to the inherent opacity of RF as a "black box" algorithm, we employed XAI for enhanced interpretability. Opting for a post-hoc approach, we utilized *SHAP*, a method for explaining individual predictions based on Shapley values. We selected the Python library *SHAP*, and we employed *explainerdashboard* which builds upon *SHAP* to offer the option of visually represent SHAP values through dashboards.

In Figure 1 we can see the *SHAP* values and how they contribute to an attempt's prediction in the model. Starting from the initial prediction percentage (50%), the corresponding *SHAP* values are added or subtracted until reaching the final prediction percentage (e.g., a *SHAP* value of 0.025 translates to a 2.5% increase in the final prediction). Green bars represent positive contributions, while red bars represent negative contributions. The most relevant features in this prediction are related to the puzzle, since it is the most challenging one in the game. Additionally, the user has a high average number of attempts per completed puzzle (`attempts_per_puzzle`), which also has a significant negative impact on the prediction. Consequently, the model predicts a success percentage of only 11.03%, accurately predicting the user's failure to solve the puzzle.

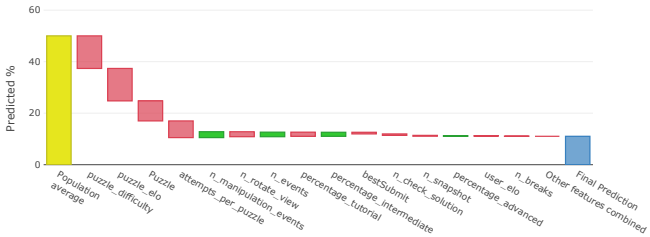


**Figure 1: SHAP values**

This type of analysis enable instructors to intervene in the classroom when necessary, identifying students who are having difficulties and helping them in real-time. Moreover, this information could also be used as input in adaptive platforms to adjust the game difficulty depending on the prediction, provide hints, and adjust the learning experience to each student's needs. By incorporating XAI techniques on top of our models, we increase the transparency of our predictive models in educational contexts, allowing instructors to understand the reasoning behind the predictions and building confidence in the potential use of SGs in the classroom.

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